

Roberto Olvera-Hernandez

Ph.D. student — Biomedical Sciences & Computational Genomics

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in [rob-olvhez](#) 🌐 [robolvhez](#)

🎓 Academic career

- Feb. 2025 - ongoing **Ph.D. in the Doctoral Program of Biomedical Sciences (PDCB)**
Center for Genomic Sciences (CCG) — UNAM ([Sohail Lab](#)) Cuernavaca, Morelos, MX.
▪ **Research project:** “Genome-wide Association Studies free of population confounds for complex traits and diseases for over 100 thousand Mexicans”
▪ **Supervisor:** Dr. Mashaal Sohail
▪ **Collaborators:** Dr. Jason Torres, Dr. Alexander T. Young
- 2018 - 2023 **B.Sc. in Biotechnology Engineering**
Instituto Tecnológico y de Estudios Superiores de Occidente (ITESO) San Pedro Tlaquepaque, Jalisco, MX.

Research placements

- Aug. 2025 - Jan. 2026 **MCPS — Research Placement**
Nuffield Department of Public Health (NDPH) — University of Oxford Oxford, Oxfordshire, U.K.
Developed a robust within-family GWAS pipeline for MCPS, enabling unbiased heritability and genetic correlation estimates for 15 complex traits—directly addressing confounding from population structure and shared environment in a highly admixed cohort.
▪ **Supervisors:** Dr. Jason Torres, Dr. Diego Aguilar-Ramirez
- Aug. 2023 - Dec. 2023 **Bachelor thesis**
Institute of Biotechnology (IBT) — UNAM Cuernavaca, Morelos, MX.
▪ **Thesis title:** “Genome-Wide Association Study (GWAS) for the identification of genes related to hydrotropic response in native maize (*Zea mays* L.) from Mexico.”
▪ **Supervisor:** Dr. Gladys Iliana Cassab López

Meetings and Conferences

- May. 2023 **XX Congreso Nacional de Biotecnología y Bioingeniería**
Mexican Society of Biotechnology and Bioengineering (SMBB)
▪ **Short communication:** <https://smbb.mx/revista-biotecnologia-2023-vol-27-no-4-2/>
- Nov. 2021 **BerryVax (ITESO) - Online Jamboree**
iGEM Design League
▪ **Project presentation:** “Use of *Bacillus subtilis* as biological machine for easy detection and elimination of plagues in berries”
▪ **Awards:** Silver Medal · Best Presentation

🧰 Professional career

- 2024 - 2025 **Income Analyst**
TOKA Internacional Zapopan, Jalisco, MX.
Income monitoring and diagnosis within the *fintech* market · Development of SQL/R scripts for massive data analysis · Generation and updating Power BI productive dashboards · Mapping and standardization of management processes such as billing.
- 2021 - 2022 **Animal Health Diagnostics Trainee**
Boehringer Ingelheim — Vetmedica Zapopan, Jalisco, MX.
NGS protocols for animal viral surveillance · Implementation of Power BI for the marketing team · Document automation using AI · Serological and molecular biology diagnostics · Lab supply management

2020 - 2021

Project Leader - International Competition

iGEM Design League 2021 (Latin America)

Awarded Silver Medal and Best Project Presentation · Project at the Latin American level · Interviews with farmers and entrepreneurs for market strategies · Training of the technical team in Synthetic Biology

Languages

Spanish (native)

Writing

Reading

Conversation



English (TOEFL iBT - C1)

Writing

Reading

Conversation



Courses and certifications

Feb. 2025 -
May. 2025

Ph.D. Selected Topic — “Clinical, Genetic, and Molecular Aspects of Obesity, Type 2 Diabetes, and Metabolic Syndrome. From the Clinic to Omics Sciences”

Unidad de Investigación Médica en Bioquímica. Unidad Médica de Alta Especialidad “Dr. Bernardo Sepúlveda”, Centro Médico Nacional Siglo XXI, IMSS, CDMX.

Jul. 2023

Course - “Bioinformatic Tools for High-Throughput Sequencing”

USMBB - IBt UNAM

Teacher Assistant in the “Programming with R”workshop.

Jul. 2021

Course - “Synthetic Biology and Genetic Circuits 101”

Scintia

Feb. 2021

Workshop - “Malice Analysis for your Project”

Engineering Biology Research Consortium (EBRC)

Feb. 2021

Course - “Agile Process, Project, and Program Controls”

Engineering Biology Research Consortium (EBRC)

Aug. 2020

Data Science: R Basics

Engineering Biology Research Consortium (EBRC)